12 Understand that cardiac muscle is myogenic and describe the normal electrical activity of the heart, including the roles of the sinoatrial node (SAN), the atrioventricular node (AVN) and the bundle of His, and how the use of electrocardiograms (ECGs) can aid the diagnosis of cardiovascular disease (CVD) and other heart conditions.

13 Explain how variations in ventilation and cardiac output enable rapid delivery of oxygen to tissues and the removal of carbon dioxide from them, including how the heart rate and ventilation rate are controlled and the roles of the cardiovascular control centre and the ventilation centre.

14 **Describe how to investigate the effects of exercise on tidal volume and breathing rate using data from spirometer traces.**

2 Describe the structure of a muscle fibre and explain the structural and physiological differences between fast and slow twitch muscle fibres.

15 Explain the principle of negative feedback in maintaining systems within narrow limits.

16 Discuss the concept of homeostasis and its importance in maintaining the body in a state of dynamic equilibrium during exercise, including the role of the hypothalamus and the mechanisms of thermoregulation.

18 Analyse and interpret data on possible disadvantages of exercising too much (wear and tear on joints, suppression of the immune system) and exercising too little (increased risk of obesity, coronary heart disease (CHD) and diabetes), recognising correlation and causal relationships.

19 Explain how medical technology, including the use of keyhole surgery and prostheses, is enabling those with injuries and disabilities to participate in sports, e.g. cruciate ligaments repair using keyhole surgery and knee joint replacement using prosthetics.

20 Outline two ethical positions relating to whether the use of performance-enhancing substances by athletes is acceptable.

17 Explain how genes can be switched on and off by DNA transcription factors including hormones.

8.4 Topic 8: Grey matter

Students will be assessed on their ability to:

1 Demonstrate knowledge and understanding of the *How Science Works* areas listed in the table on page 12 of this specification.

3 Describe the structure and function of sensory, relay and motor neurones including the role of Schwann cells and myelination.

7 Explain how the nervous systems of organisms can cause effectors to respond as exemplified by pupil dilation and contraction.

4 Describe how a nerve impulse (action potential) is conducted along an axon including changes in membrane permeability to sodium and potassium ions and the role of the nodes of Ranvier.

2 Describe how plants detect light using photoreceptors and how they respond to environmental cues.

5 Describe the structure and function of synapses, including the role of neurotransmitters, such as acetylcholine.

6 Describe how the nervous systems of organisms can detect stimuli with reference to rods in the retina of mammals, the roles of rhodopsin, opsin, retinal, sodium ions, cation channels and hyperpolarisation of rod cells in forming action potentials in the optic neurones.

8 Compare mechanisms of coordination in plants and animals, i.e. nervous and hormonal, including the role of IAA in phototropism (details of individual mammalian hormones are not required).

9 Locate and state the functions of the regions of the human brain's cerebral hemispheres (ability to see, think, learn and feel emotions), hypothalamus (thermoregulate), cerebellum (coordinate movement) and medulla oblongata (control the heartbeat).

10 Describe the use of magnetic resonance imaging (MRI), functional magnetic resonance imaging (fMRI) and computed tomography (CT) scans in medical diagnosis and investigating brain structure and function.

11 Discuss whether there exists a critical ‘window’ within which humans must be exposed to particular stimuli if they are to develop their visual capacities to the full.

14 Describe how animals, including humans, can learn by habituation.

12 Describe the role animal models have played in developing explanations of human brain development and function, including Hubel and Wiesel’s experiments with monkeys and kittens.

16 Discuss the moral and ethical issues relating to the use of animals in medical research from two ethical standpoints.

17 Explain how imbalances in certain, naturally occurring, brain chemicals can contribute to ill health (e.g. dopamine in Parkinson’s disease and serotonin in depression) and to the development of new drugs.

18 Explain the effects of drugs on synaptic transmissions, including the use of L-Dopa in the treatment of Parkinson’s disease and the action of MDMA in ecstasy.

19 Discuss how the outcomes of the Human Genome Project are being used in the development of new drugs and the social, moral and ethical issues this raises.

20 Describe how drugs can be produced using genetically modified organisms (plants and animals and micro organisms).

21 Discuss the risks and benefits associated with the use of genetically modified organisms.

13 Consider the methods used to compare the contributions of nature and nurture to brain development, including evidence from the abilities of newborn babies, animal experiments, studies of individuals with damaged brain areas, twin studies and cross-cultural studies.

15 **Describe how to investigate habituation to a stimulus**

Generic units (concept and context)

The following section contains details of the external assessments for Units 3 and 6.
The same external assessments are used for both the concept-led and context-led approaches.